

Peng Jiang

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Education

Tsinghua University	School of Life Sciences	Bachelor's Degree	2017.08–2021.06
Tsinghua University	Department of Computer Science and Technology	Undergraduate Minor	2018.08–2021.06
Tsinghua University	School of Life Sciences	PhD Student in Computational Neuroscience	2021.08–present
Supervisor: Xiaoxuan Jia			

Research Interests

Keywords: Neural Foundation Model; Neural Encoding & Decoding; World Model; Neural Representation.

Current focus: Neural foundation models, exploring how to build neural pre-trained models for neural encoding and decoding research.

Academic Experience

Multi-task Neural Representation Research 2022–2024

Neural Coding Lab Tsinghua University

- Analyzed multi-task neural representations across brain regions using invasive Neuropixels recordings.
- Built multi-area RNN models to study hierarchical task representation in flexible cognition.
- Studied compositional generalization via continual learning with LoRA fine-tuning of LLMs.

NeuroHorizon: Long-Horizon Forward Prediction of Neural Population Activity via Autoregressive Decoding with Hierarchical Memory 2026

Peng Jiang, Xiaoxuan Jia · Submitted to NeurIPS 2026 · Under review*

- Studied long-horizon forward prediction of neural population activity.
- Developed an autoregressive encoder-decoder with spike tokenization and hierarchical memory.
- Evaluated multi-step rollout stability across motor- and visual-cortex datasets.

Project Experience

Real-time audio-driven digital human based on 3DGS 2024–2025

Co-Founder and Algorithm Lead Startup · Leave of absence from PhD

- Built a real-time, audio-driven interactive digital human system based on 3D Gaussian Splatting (3DGS).
- Maintained the pipeline for capture calibration, 3DGS reconstruction, and audio-to-expression driving.

NeuroGPT project 2023.06

Tsinghua University LLM Application Design Challenge · 4th place / 94 teams

- Used multimodal large language models as an information relay, decoding neural semantics and mapping them into action/control signals for medical, educational, and entertainment scenarios.